

Abstract

The aim of our experiment was to identify whether the quoted heat transfer rates from the manufacturers of the Hilton A660 AC unit were accurate. To accomplish this, measurements of volume of condensate collected and orifice pressure via a manometer were taken over a 3-minute period. Additionally, temperature readings at 4 distinct stages of the AC unit (from dry and wet bulb thermometers) were also taken. This process was repeated 10 times. Using a psychometric chart calculator, steady flow energy equations and mass conservation equations, the experimental values for heat transfer rates were obtained ($\dot{Q}_c$, $\dot{Q}_{rh}$, $\dot{Q}_{ph}$ and $\dot{Q}_b$). Additional assumptions were used to simplify these equations, such as assuming negligible environmental drift. Whilst the quoted value of $\dot{Q}_{ph} + \dot{Q}_b = 6 \text{ kW}$ is within the uncertainty range of our experimental value $5.899 \pm 0.4846 \text{ kW}$ (with a minimal error of -1.68%), there is an exceedingly large error for $\dot{Q}_{rh}$ at -82.2% (experimental value of $0.178 \pm 0.464 \text{ kW}$ compared to the quoted value of 1 kW). This error is reinforced through observation of a plotted psychometric process of stages 1 to 4, where process 3-4 has a steep diagonal line that signifies major dehumidifying (when it should be purely heating and yield a horizontal process line). This is likely due to issues with our boiler leaking steam, leading to anomalous results for 5 out of our 10 repeats. Moreover, only one point on the cross-sectional area was measured when temperature distribution would not be uniform.

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Nomenclature

ω	Specific Humidity
ϕ	Relative Humidity
h	Mixture Specific Enthalpy
v	Specific Volume
$\dot{Q}_b$	Boiler Heat Transfer rate
$\dot{Q}_c$	Cooler Heat Transfer Rate
$\dot{Q}_{rh}$	Re-Heater Heat Transfer Rate
$\dot{Q}_{ph}$	Pre-Heater Transfer Rate
$\dot{W}_F$	Fan Work Rate
$\dot{m}_a$	Mass Flowrate of Air
$\dot{m}_v$	Mass Flowrate of Vapour
$\dot{m}_{con}$	Mass Flowrate of Condensate
$\dot{m}_W$	Mass Flowrate of Water
P	Pressure
ρ	Density
g	Acceleration of Gravity
z	Manometer height
$\dot{V}$	Volumetric Flowrate
V	Volume
t	Time
T_{DB}	Dry Bulb Temperature
T_{WB}	Wet Bulb Temperature

1. Introduction

The aim of our experiment is to calculate the experimental values for mass flow rates and heat transfer rates ($\dot{Q}_c$, $\dot{Q}_{rh}$, $\dot{Q}_{ph}$ and $\dot{Q}_b$) at four distinct stages of an AC unit (Figure 1). After this, we compare it to the quoted values provided by the manufacturer and judge whether our experimental values align with them.



Figure 1: Hilton A660 AC unit [1]

The experimental values are calculated using a psychometric chart calculator, obtaining values for ϕ , ω , v and h . Thermodynamic analysis, including the use of SFEE and MCE equations, are subsequently used with the aid of appropriate assumptions to simplify these equations.

2.1 Experimental Procedure

The experiment used a Hilton A660 air conditioning unit (schematic as Figure 2) with 4 distinct stages:

1. Steam generated from an electric heater boiler is injected into the airstream
2. An electrical preheater raises the temperature
3. A cooler removes some of the moisture through condensation
4. An electrical reheater to raise the air temperature again

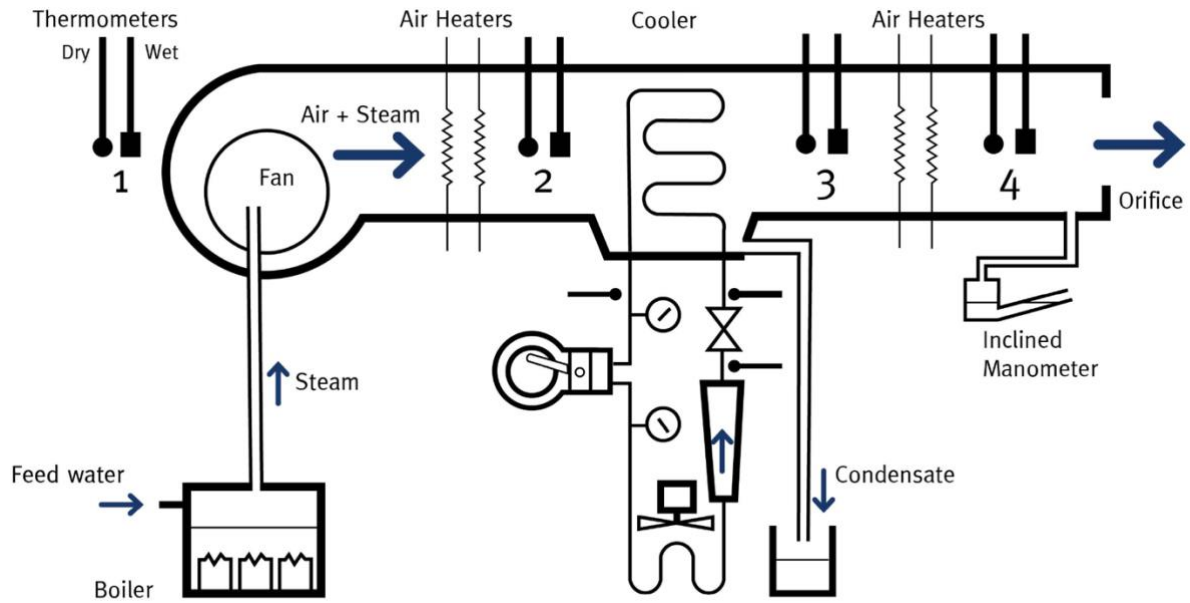


Figure 2: Diagram of AC unit with labelled stages [2]

Moreover, the AC unit being used was operating at condition 2 (shown in Table 1).

Table 1: Values for AC unit being used at condition 2

$\dot{Q}_b$	5 kW
$\dot{Q}_{ph}$	1 kW
$\dot{Q}_{rh}$	1 kW
V_f	190 V

Two thermometers, one measuring the dry bulb and the other measuring the wet bulb temperature, were placed at each stage for measurements. Also, any condensate that leaks out via an outlet pipe is collected in a beaker whilst an inclined manometer measures the pressure drop as the air-vapour mixture flows through the orifice.

Two temperature values (dry and wet bulb) were measured as those are all we need to plot a thermodynamic process on a psychometric chart and obtain values needed solve heat transfer rates.

Prior to the experiment starting, the initial room temperature and pressure was measured (as shown in Table 1).

Table 1: Initial room temperature and pressure readings

Room Temperature ($^{\circ}\text{C}$)	Room Pressure (mbar)
24.0	1017.0

With a measurement uncertainty of $\pm 0.5^{\circ}\text{C}$ for the thermometer and $\pm 0.5\text{mm}$ for the manometer. Additionally, we checked whether the 4 dry bulb thermometers were giving the same reading as the room temperature measured above - confirming initial conditions of AC unit were equivalent to those of the room.

Then, we started the experiment and, every 3 minutes, took readings of all 4 stages dry and wet bulb temperatures using thermometers and the pressure using the inclined manometer. During the 3 minutes between readings, the volume of condensate collected (over 3 minutes) was measured via a measuring jug. This process was repeated 10 times.

2.2 Data Analysis

The values obtained during our experiment were recorded in Table 2 below (where $T_{\text{odd}} = T_{\text{DB}}$ and $T_{\text{even}} = T_{\text{WB}}$).

Table 2: Raw temperature, volume and pressure data measured

Reading	Stage 1		Stage 2		Stage 3		Stage 4		V (mL)	Mano. (mm H_2O)
	$T_1(^{\circ}\text{C})$	$T_2(^{\circ}\text{C})$	$T_3(^{\circ}\text{C})$	$T_4(^{\circ}\text{C})$	$T_5(^{\circ}\text{C})$	$T_6(^{\circ}\text{C})$	$T_7(^{\circ}\text{C})$	$T_8(^{\circ}\text{C})$		
1	28.0	19.0	35.0	23.5	23.0	20.0	31.0	22.0	38.5	12.4
2	28.5	19.75	35.0	25.0	22.5	19.0	30.0	21.0	18.0	12.3
3	28.5	20.00	35.5	23.0	22.5	19.0	30.0	21.0	14.0	12.4
4	N/A									
5										
6	29.5	20.5	34.5	28.0	27.0	25.0	34.5	21.5	47.5	12.2
7	29.5	20.5	37.5	28.5	27.5	25.5	35.0	26.5	49.5	12.3
8	30.0	21.0	38.0	28.0	27.0	25.0	35.0	26.0	58.5	12.4
9	30.0	21.0	38.0	29.0	27.5	25.5	35.0	27.0	55.5	12.2
10	29.0	20.5	37.0	28.0	27.0	25.0	34.5	26.5	64.5	12.8
Avg.	29.6	20.7	37.6	28.3	27.2	25.2	34.8	26.5	55.1	12.4
% Unc.	1.69%	1.21%	4.65%	1.77%	0.92%	0.99%	2.16%	1.89%	15.4%	2.42%

During readings 1-3 of our experiment, we observed that our boiler was letting off a significant amount of steam and therefore efficiency of the first stage was considerably low, affecting all our other stages as energy provided (from boiler) is not equal to energy input of the AC unit. So, readings 1-3 are considered anomalous. Additionally, reading 6 of $T_8^\circ\text{C}$ varies greatly from other $T_8^\circ\text{C}$ readings and thus is also considered anomalous.

Moreover, we only had time for 5 readings after the boiler was fixed and so they are no readings for 4 and 5. Hence, only readings 6-10 were considered to calculate the average values for temperature, volume collected and pressure (using Equation 1):

$$Average = \frac{\sum x}{n} \quad (1)$$

To find the uncertainties of these values, we used the range method to find the absolute uncertainty (Equation 2) and then the subsequent percentage uncertainties (Equation 3)[3]:

$$Unc. = \frac{x_{max} - x_{min}}{2} \quad (2)$$

$$\% Unc. = \frac{Unc.}{Average} \quad (3)$$

Where an example of these calculations is done for T_1 in Appendix A.

Using an online psychrometric chart calculator [4] the 4 pairs of T_{DB} and T_{WB} average values were inputted to obtain values for ω, ϕ, v (m^3 / kg) and h (kJ / kg) for all 4 stages (as shown in Table 3 below).

Table 3: Psychrometric calculator values for ω, ϕ, v (m^3/kg) and h (kJ/kg)

	ω	ϕ	v (m^3/kg)	h (kJ/kg)
1	0.0116	0.45	0.874	59.51
2	0.0206	0.50	0.909	90.71
3	0.0195	0.85	0.878	77.06
4	0.0168	0.48	0.896	78.00

The online calculator was used as there is minimal uncertainty in obtaining values compared to reading them off the chart itself (with uncertainty derived from reading values that lie between line intervals of the chart).

To find the percentage uncertainties of these psychrometric values, the lower bound temperature values were inputted to find the x_{min} , upper bound temperature values to find the upper bound x_{max} , and hence used in Equations 2 and 3.

An example of these calculations for ω_1 is done in Appendix B, and all the percentage uncertainty values are shown in Table 4 below.

Table 4: Psychrometric calculator % uncertainty values

	ω	ϕ	v (m^3/kg)	h (kJ/kg)
1	1.29%	1.11%	0.17%	1.45%
2	0.97%	3.00%	0.61%	2.61%
3	1.79%	0.00%	0.11%	1.37%
4	5.17%	1.04%	0.33%	2.84%

Using the values obtained, we plotted a thermodynamic process of our 4 stages on a psychrometric chart (Figure 3 below) for stages 1-4.

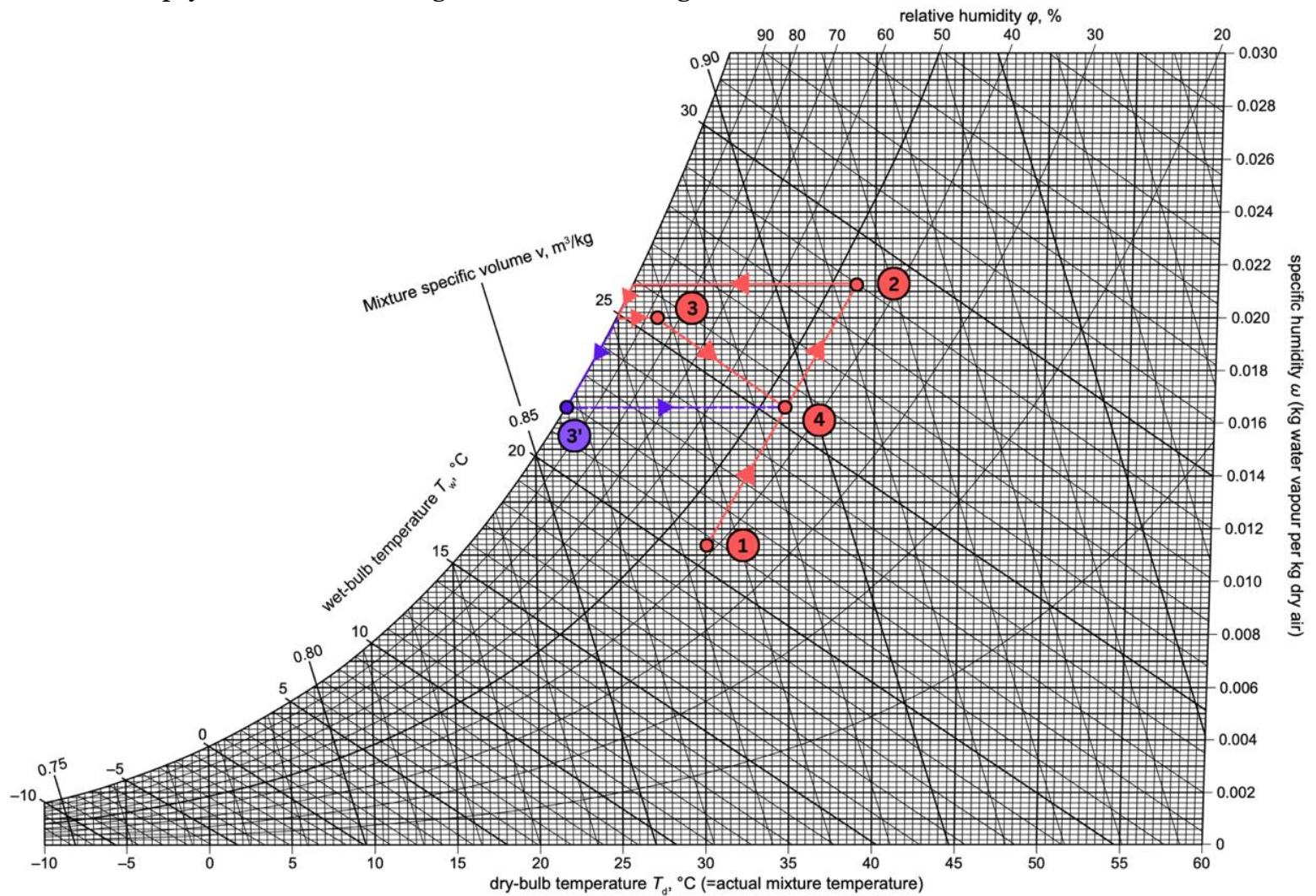


Figure 3: Psychrometric chart annotated with 4 experimentally obtained stages (and 1 theoretical stage 3') [5]

On the chart, there is a large disparity between the position of experimental stage 3 and the theoretical stage 3'. This is because 3' assumes purely dehumidifying and cooling for process 2-3' and purely reheating for process 3'-4 (and therefore $\omega_{3'} = \omega_4$).

But process 2-3 exhibits significantly less cooling (as $T_3 > T_{3'}$) and less dehumidifying ($\omega_3 > \omega_{3'}$).

Therefore, it can be perceived that the cooling system for the AC unit operated at a lower efficiency than the theoretical model, as it failed to reach the saturation state required for maximum dehumidification and cooling.

2.3 Heat Transfer Rates

To calculate the heat transfer rates from the electrical components, must calculate values for processes in reverse, starting at process 3-4 and ending in process 1-2, with a diagram (Figure 4) of mass flow rates, heat transfer rates and power inputs to be calculated below.

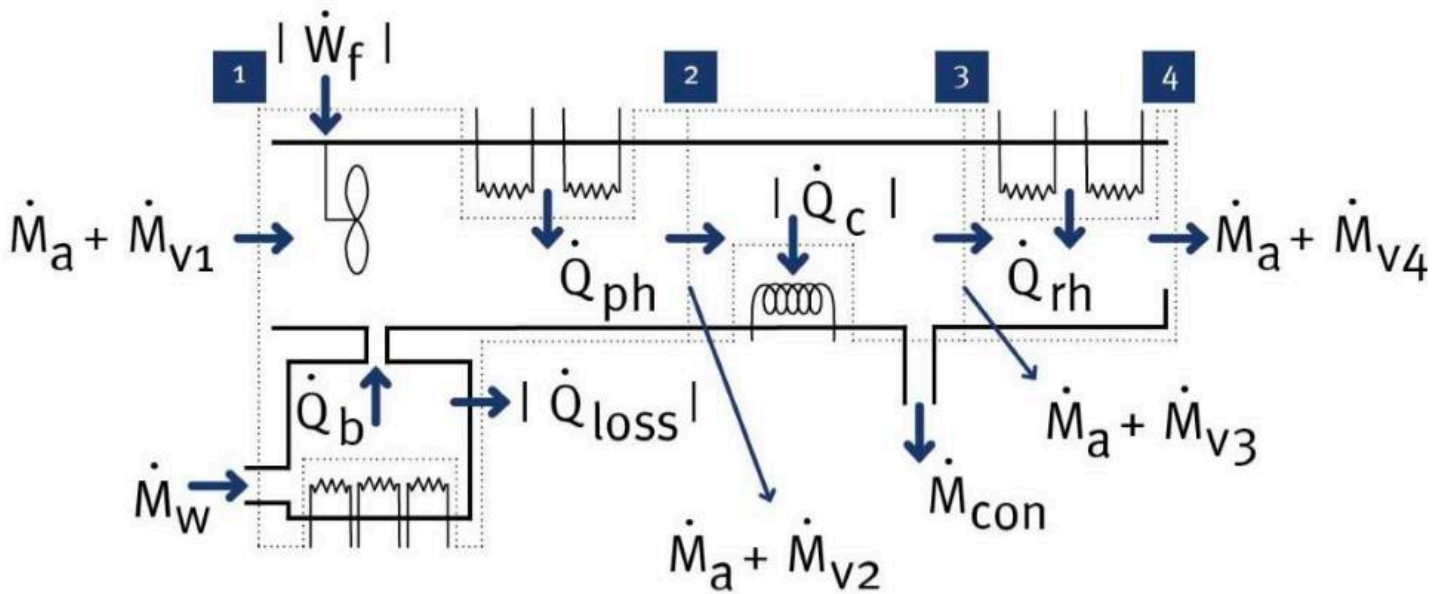


Figure 4: AC unit with labelled mass flow rates, heat transfer rates, and power inputs [1]

2.3.1 Process 3-4 (Re-heater):

Using v_4 and the average manometer height in $mm H_2O$, the mass flow rate at stage 4 $\dot{m}_4$ can be calculated using Equation 4 below:

$$\dot{m}_4 = 0.0517 \sqrt{\frac{L}{v_4}} \quad (4)$$

$$\dot{m}_4 = 0.0517 \sqrt{\frac{12.4}{0.896}} = 0.1923 \text{ kg s}^{-1}$$

Moreover, the absolute uncertainty values are found using partial differentiation, where the general formula is shown below (Equation 5)[6]:

$$\delta y = \sqrt{\left(\frac{\partial f}{\partial x_1} \delta x_1\right)^2 + \left(\frac{\partial f}{\partial x_2} \delta x_2\right)^2 + \dots + \left(\frac{\partial f}{\partial x_n} \delta x_n\right)^2}, \text{ where } y = f(x_1, x_2 \dots x_n) \quad (5)$$

An uncertainty for $\dot{m}_4$ is found (with an example calculation shown below as equations 6, 7 and 8):

$$\frac{\delta \dot{m}_4}{\dot{m}_4} = \frac{\sqrt{\left(\frac{\partial \dot{m}_4}{\partial L} \delta L\right)^2 + \left(\frac{\partial \dot{m}_4}{\partial v_4} \delta v_4\right)^2}}{\dot{m}_4} \quad (6)$$

$$\begin{aligned} \frac{\partial \dot{m}_4}{\partial L} &= \frac{1}{2} * 0.0517 * L^{-\frac{1}{2}} * v_4^{-\frac{1}{2}} \\ \frac{\partial \dot{m}_4}{\partial L} &= \frac{1}{2} * 0.0517 * (12.4)^{-\frac{1}{2}} * (0.896)^{-\frac{1}{2}} = 0.00776 \end{aligned} \quad (7)$$

$$\begin{aligned} \frac{\partial \dot{m}_4}{\partial v_4} &= -\frac{1}{2} * 0.0517 * L^{\frac{1}{2}} * v_4^{-\frac{3}{2}} \\ \frac{\partial \dot{m}_4}{\partial v_4} &= -\frac{1}{2} * 0.0517 * (12.4)^{\frac{1}{2}} * (0.896)^{-\frac{3}{2}} = -0.10733 \end{aligned} \quad (8)$$

$$\frac{\delta \dot{m}_4}{\dot{m}_4} = \frac{\sqrt{(0.00776 * 0.3)^2 + (-0.10733 * 0.00296)^2}}{0.19233} = 1.22\% \quad (6)$$

$$\delta \dot{m}_4 = 0.002348$$

Therefore, $\dot{m}_4 = 0.1923 \pm 0.002348 \text{ kg/s}$.

Moving forward, absolute uncertainty values (not including steam table values) will be calculated using the same method in above, with working out displayed in the relevant sections of the appendix.

The expected rate of heat loss from the boiler directly to the atmosphere, or the rate of cooler heat transfer $|\dot{Q}_c|$, can also be found using the calculated ambient temperature T (with a measurement uncertainty of $\pm 0.5^\circ\text{C}$) as Equation 9 below, with uncertainty found in Appendix C calculations:

$$|\dot{Q}_c| \text{ (in W)} = 4.3 (100 - T) \quad (9)$$

$$|\dot{Q}_c| = 4.3 (100 - 24) = 326.8 \pm 2.2 \text{ W}$$

The mass conservation equation (MCE) for process 3-4 can be applied (Equation 10):

$$\dot{m}_{a3} + \dot{m}_{v3} = \dot{m}_{a4} + \dot{m}_{v4} \quad (10)$$

It can be perceived above that there is only dry air and water vapour, and as a purely heating process assumed (so no moisture input, $\dot{m}_{v3} = \dot{m}_{v4} = \dot{m}_v$), it can be deduced that $\dot{m}_{a3} = \dot{m}_{a4} = \dot{m}_a$.

The SFEE (where no work input and only reheat addition $\dot{Q}_{rh}$) can then be simplified using the relations obtained as Equation 11 below:

$$\dot{Q}_{rh} = \dot{m}_a(h_{a4} - h_{a3}) + \dot{m}_v(h_{v4} - h_{v3}) \quad (11)$$

$$\dot{Q}_{rh} = \dot{m}_a(h_4 - h_3)$$

And according to the equation for specific humidity (Equation 12):

$$\omega_4 = \frac{\dot{m}_v}{\dot{m}_a} = \frac{\dot{m}_4 - \dot{m}_a}{\dot{m}_a}, \text{ where } \dot{m}_v + \dot{m}_a = \dot{m}_4 \quad (12)$$

$$\dot{m}_a = \frac{\dot{m}_4}{1 + \omega_4}$$

After substituting back into Equation 11, Equation 13 can be obtained, with uncertainty found in Appendix D calculations:

$$\dot{Q}_{rh} = \frac{\dot{m}_4}{1 + \omega_4} (h_4 - h_3) \quad (13)$$

$$\dot{Q}_{rh} = \frac{0.1923}{1 + 0.0168} (78 - 77.06) = 0.178 \pm 0.464 \text{ kW}$$

Leading to the rate of re-heat transfer being found. Moreover, $\dot{Q}_{rh}$ is a positive value as heat is being added to the system rather than rejected.

2.3.2 Process 2-3 (Cooler and Dehumidifier):

The MCE for 2-3 (as Equation 14) has an additional supply of air at stage 3 due to the exit of condensate from the AC unit. This equation can be simplified as air is neither added nor removed, therefore $\dot{m}_{a2} = \dot{m}_{a3}$.

$$\dot{m}_{a2} + \dot{m}_{v2} = \dot{m}_{a3} + \dot{m}_{v3} + \dot{m}_{con} \quad (14)$$

$$\dot{m}_{v2} = \dot{m}_{v3} + \dot{m}_{con}$$

Then, as $\dot{m}_{con}$ is proportional to the volumetric flow rate $\dot{V}$ (of the volume of water collected in the beaker), we can use Equation 15 to find a value.

$$\dot{m}_{con} = \frac{\dot{V}}{v_{con}} = \frac{V \text{ (in } m^3\text{)}}{v_{con} * t \text{ (in } s\text{)}} \quad (15)$$

To find v_{con} , we use the saturated water and steam tables at point $T_{con} = 25.0^\circ C$, where the condensate is assumed to be a saturated liquid. Hence, $v_f = v_{con} = 0.001003 \text{ m}^3/kg$.

Also, for steam table values, their uncertainties are found by finding the difference between the respective values one above and one below and halving to find the absolute uncertainty (essentially the range method described before).

Hence, an uncertainty for v_{con} is found using equations 2, 16 and 17 below, where temperature taken as $25^\circ C$ so must look at $24^\circ C$ and $26^\circ C$:

$$\text{At } 24^\circ C: v_f = 0.001003 \text{ m}^3/kg \quad (16)$$

$$\text{At } 26^\circ C: v_f = 0.001003 \text{ m}^3/kg \quad (17)$$

$$Unc. = \frac{0.001003 - 0.001003}{2} = \pm 0 \text{ m}^3/kg \quad (2)$$

And therefore, $v_{con} = 0.001003 \pm 0 \text{ m}^3/kg$.

In future, uncertainty values obtained by the range method will also have working shown in relevant sections of the appendix.

Using the values obtained: $\dot{m}_{con} = \frac{55.1 \cdot 10^{-6}}{(0.001003)(3 \cdot 60)} = 0.0003052 \pm 0.00004700 \text{ kg/s}$, with uncertainty found in Appendix E calculations.

The value for time t was considered to have a negligible uncertainty. This is because the error due to human reaction time is roughly $\pm 0.2s$ [7] which is insignificant compared to the 180s reading taken.

Using Equation 12, we deduce that $\dot{m}_{v3} = \dot{m}_a \omega_3$ and $\dot{m}_{v2} = \dot{m}_a \omega_2$ according to the specific humidity relation.

Furthermore, substituting the derived $\dot{m}_a$ equation (from Equation 14) into this deduced relation, we can obtain an equation for $\dot{m}_{v3}$ (Equation 18), with uncertainty found in Appendix F calculations:

$$\dot{m}_{v3} = \frac{\dot{m}}{1 + \omega_3} \omega_3 \quad (18)$$

$$\dot{m}_{v3} = \frac{0.1923}{1 + 0.0195} * 0.0195 = 0.00368 \pm 0.000079 \text{ kg/s}$$

Moreover, we can now solve for $\dot{m}_{v2}$ using Equation 14 and uncertainty calculations in Appendix G:

$$\dot{m}_{v2} = \dot{m}_{v3} + \dot{m}_{con} \quad (14)$$

$$\dot{m}_{v2} = 0.00368 + 0.0003052 = 0.0039852 \pm 0.00009192 \text{ kg/s}$$

However, there is a second method of finding $\dot{m}_{v2}$ using the specific humidity relations directly (with uncertainty calculations also in Appendix G)

Similar to Equation 18, can obtain a relation for $\dot{m}_{v2}$ in terms of ω_2 (Equation 19):

$$\dot{m}_{v2} = \frac{\dot{m}}{1 + \omega_2} \omega_2 \quad (19)$$

$$\dot{m}_{v2} = \frac{0.1923}{1 + 0.0206} * 0.0206 = 0.00388 \pm 0.0000601 \text{ kg/s}$$

As method 2 has a lower uncertainty ($0.00009192 > 0.0000601$) than method 1, we will go forward using $\dot{m}_{v2} = 0.00388 \pm 0.0000601 \text{ kg/s}$.

Now, writing out the SFEE equation (where no work and only heat input $\dot{Q}_c$) as Equation 20 below:

$$\dot{Q}_c = \dot{m}_a(h_{a3} - h_{a2}) + \dot{m}_{v3}h_{v3} - \dot{m}_{v3}h_{v2} + \dot{m}_{con}h_{con} \quad (20)$$

$$\dot{Q}_c = \dot{m}_a(h_3 - h_2) + \dot{m}_{con}h_{con}, \text{ as } \dot{m}_a(h_3 - h_2) = \dot{m}_a(h_{a3} - h_{a2}) + \dot{m}_{v3}h_{v3} - \dot{m}_{v3}h_{v2}$$

The value for $\dot{m}_a$ can be found using Equation 21, with uncertainty found in Appendix H calculations:

$$\dot{m}_a = \frac{\dot{m}}{1 + \omega_3} \quad (21)$$

$$\dot{m}_a = \frac{0.1923}{1 + 0.0195} = 0.1886 \pm 0.002304 \text{ kg/s}$$

Also, the value for h_{con} is found using the saturated water and steam tables, where at $T = 25^\circ\text{C}$ and $h_{con} = h_f = 104.8 \text{ kJ/kg}$ (as condensate is a fully saturated liquid).

Additionally, using the uncertainty calculations from Appendix I, we find that:

$$h_{con} = 104.8 \pm 4.15 \text{ kJ/kg}$$

Thus, the SFEE can be solved, with uncertainty found in Appendix J calculations:

$$\dot{Q}_c = \dot{m}_a(h_3 - h_2) + \dot{m}_{con}h_{con} \quad (20)$$

$$\dot{Q}_c = 0.1886(77 - 90.71) + 0.0003052 * 104.8 = -2.54 \text{ kW}$$

$$|\dot{Q}_c| = 2.54 \pm 0.490 \text{ kW}$$

2.3.3 Process 1-2 (Humidifier and Preheater):

The MCE for process 1-2 is shown below as Equation 22, where the $\dot{m}_w$ term is the mass flow rate from steam supplied by the boiler.

$$\dot{m}_w + \dot{m}_{a1} + \dot{m}_{v1} = \dot{m}_{a2} + \dot{m}_{v2} \quad (22)$$

$$\dot{m}_{a1} = \dot{m}_{a2} = \dot{m}_a$$

$$\dot{m}_w + \dot{m}_{v1} = \dot{m}_{v2}$$

For the SFEE (Equation 23), they are both heat transfer and power terms. For heat transfer, they are heat additions from the boiler $\dot{Q}_b$ and the air heaters $\dot{Q}_{ph}$ and heat rejections from the condensate $\dot{Q}_{loss}$. There is also energy removed by the fan $-\dot{W}_f$.

$$\dot{Q}_{ph} + \dot{Q}_b - \dot{Q}_{loss} + \dot{W}_f = \dot{m}_a(h_{a2} - h_{a1}) + \dot{m}_{v2}h_{v2} - \dot{m}_{v1}h_{v1} - \dot{m}_wh_w \quad (23)$$

$$\dot{Q}_{ph} + \dot{Q}_b - \dot{Q}_{loss} + \dot{W}_f = \dot{m}_a(h_2 - h_1) - \dot{m}_wh_w$$

According to the AC operating at condition 2: $\dot{W}_f = 190 \text{ W}$, and using the same relation as Equation 13, we can simplify down to Equation 24:

$$\dot{Q}_{ph} + \dot{Q}_b - \dot{Q}_{loss} + \dot{W}_f = \frac{\dot{m}}{1 + \omega_1}(h_2 - h_1) - \dot{m}_wh_w \quad (24)$$

Moreover, using the relation in Equation 19, the equation can be further simplified down to Equation 25:

$$\dot{m}_{v1} = \frac{\dot{m}}{1 + \omega_1} \omega_1 \quad (25)$$

$$\dot{m}_w = \dot{m}_{v2} - \dot{m}_{v1}$$

$$\dot{Q}_{ph} + \dot{Q}_b = \frac{\dot{m}}{1 + \omega_1}(h_2 - h_1) - (\dot{m}_{v2} - \dot{m}_{v1})h_w - \dot{W}_f + \dot{Q}_{loss}$$

We assume that we operate at a feedwater temperature that is equal to the measured ambient temperature $T_w = T_{amb} = 24^\circ\text{C}$. This is because the stage 1 is the start of the AC process and therefore assume minimal temperature drift.

So, $h_w = h_f = 100.7 \pm 4.165 \text{ kJ/kg}$ (with uncertainty found in Appendix K calculations) as feedwater is assumed to be a completely saturated liquid.

Therefore, we can solve Equation 24, with uncertainty found in Appendix L calculations:

$$\dot{Q}_{ph} + \dot{Q}_b = \frac{0.1923}{1 + 0.0116} (90.71 - 59.51) - (0.00388 - 0.002205)(100.7) - (0.19) + (0.3268) \quad (24)$$

$$\dot{Q}_{ph} + \dot{Q}_b = 5.899 \pm 0.4846 \text{ kW}$$

Where the percentage errors for $\dot{Q}_{ph} + \dot{Q}_b$ (Equation 26) and $\dot{Q}_{rh}$ (Equation 27) are below.

$$\% \text{ Error } (\dot{Q}_{ph} + \dot{Q}_b) = \frac{5.899 - 6}{6} * 100 = -1.68\% \quad (26)$$

$$\% \text{ Error } (\dot{Q}_{rh}) = \frac{0.178 - 1}{1} * 100 = -82.2\% \quad (27)$$

To measure $\dot{Q}_{ph}$ and $\dot{Q}_b$ independently, a set of dry/wet bulb thermometers must be placed to measure the temperature between the pre-heater and the boiler.

Looking at the MCE over the whole unit as Equation 28:

$$\dot{m}_w + \dot{m}_{v1} + \dot{m}_a = \dot{m}_{con} + \dot{m}_{v4} + \dot{m}_a \quad (28)$$

$$\dot{m}_w + \dot{m}_{v1} = \dot{m}_{con} + \dot{m}_{v4}$$

$$\text{Mass flow rate in} = \dot{m}_w + \dot{m}_{v1} = 0.00388 \pm 0.0000601 \text{ kg/s}$$

$$\text{Mass flow rate out} = \dot{m}_{con} + \dot{m}_{v4} = 0.00348 \pm 0.0000470 \text{ kg/s}$$

3. Discussion

As can be seen in the analysis section, the error between our experimentally derived and theoretical values for $\dot{Q}_{ph} + \dot{Q}_b$ is minimal at -1.68%, where the calculated value aligns with the quoted manufacturer value (as 6 kW is within the uncertainty range of $5.899 \pm 0.4846 \text{ kW}$). However, there is an exceedingly large error for $\dot{Q}_{rh}$ at -82.2% (where even taking the maximum value of $0.178 \pm 0.464 \text{ kW}$ leads to a -35.8% error).

This large disparity is supported by Figure 3 where it clearly shows that for process 3-4, instead of purely heating leading to a perfectly horizontal line, there is a steep diagonal that signifies major dehumidifying. Whilst it can be assumed that in an experimental setting heating often leads to a small dehumidifying effect, this effect is much greater than expected - thus leading to a large error in $\dot{Q}_{rh}$.

This error could be due to many factors. Firstly, our calculations assume no environmental drift - where all temperature and humidity conditions stay constant over time. However, there is a significant uncertainty in our volume of condensate collected values due to varying measurements (15.4%), signalling that conditions continued to vary as time went on. Additionally, even after removing anomalous results, our temperature uncertainties remain high (with a maximum uncertainty of 4.65% for $T_3^\circ\text{C}$). This is likely due to our experiment only having 5 valid repeats rather than the desired 10, because of the boiler leaking steam initially. It follows our intuition that if the number of repeats was increased to 10, the uncertainty in our raw data would decrease and hence yield a more accurate value for $\dot{Q}_{rh}$.

Moreover, it is unlikely that energy is lost only in the boiler, as energy is often lost in heating coils due to poor maintenance / excessive rusting. Also, temperature readings were only taken at one point of the duct cross section, assuming uniformity. But it is unlikely that the temperature distribution would be uniform. So, more thermometers should have been positioned at varying heights, with a weighted average of all dry and wet bulb thermometers per stage likely yielding more accurate temperature values.

These errors and unsteady conditions lead to a larger error for $\dot{Q}_{rh}$ than $\dot{Q}_{ph} + \dot{Q}_b$ because the reheating process is the last process of the AC unit and so accumulates errors from the 3 prior processes.

When analysing our calculated mass flow rates, we can observe that mass conservation is not followed as the mass flow rate into the system $0.00388 \pm 0.0000601 \text{ kg/s}$ is not within the uncertainty range for mass flow rate out of the system $0.00348 \pm 0.0000470 \text{ kg/s}$ - yielding an error of 11.49%. This is most likely due to the issues with our boiler leaking steam initially (and likely persisting at a smaller degree during trials 6-10). As the actual $\dot{m}_w < \text{theoretical } \dot{m}_w$ due to steam leakage, this results in an underestimate for mass flow rate out of the system (which is the case).

4. Conclusion

Whilst our calculated value of $\dot{Q}_{ph} + \dot{Q}_b$ aligns relatively closely with the quoted value (-1.68% error), there is a significant discrepancy between our calculated $\dot{Q}_{rh}$ value and the quoted value (-82.2% error); therefore, our experiment was not able to verify that AC unit operated similar to the manufactures intentions. Additionally, we determined that the large error was derived from environmental drift and a lack of repeats (due to our AC unit not working for the first 5 trials).

Going forward, additional dry and wet bulb thermometers could be used at different depths to account for a non-uniform temperature distribution at the duct. Also, measured values should only be considered after the system is perceived to of stabilised (values are steady).

No AI was used to create the content in this report. No AI was used for guidance in preparing the content.

5. References

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6. Appendix

6.1 APPENDIX A:

Average and uncertainty calculations for T_1 :

$$Average = \frac{29.5 + 29.5 + 30.0 + 30.0 + 29.0}{5} = 29.6^{\circ}C$$

$$Unc. = \frac{30 - 29}{2} = \pm 0.5^{\circ}C$$

$$\% Unc. = \frac{0.5}{29.6} * 100\% = 1.69\%$$

6.2 APPENDIX B:

Uncertainty values for ω_1 :

Inputting lower bound value of T_1 (29.0°C) and T_2 (20.5°C) - $\omega_{1,lower} = 0.0116$

Inputting upper bound value of T_1 (30.0°C) and T_2 (21.0°C) - $\omega_{1,upper} = 0.0119$

$$Unc. = \frac{0.0119 - 0.0116}{2} = 0.00015$$

$$\% Unc. = \frac{0.00015}{0.0116} * 100\% = 1.29\%$$

6.3 APPENDIX C:

Uncertainty calculations for $|\dot{Q}_{loss}|$:

$$\frac{\delta|\dot{Q}_{loss}|}{|\dot{Q}_{loss}|} = \frac{\sqrt{\left(\frac{\partial|\dot{Q}_{loss}|}{\partial T} \delta T\right)^2}}{|\dot{Q}_{loss}|}$$

$$\frac{\partial|\dot{Q}_{loss}|}{\partial T} = -4.3$$

$$\frac{\delta|\dot{Q}_{loss}|}{|\dot{Q}_{loss}|} = \frac{\sqrt{(-4.3 * 0.5)^2}}{326.8}$$

$$\delta|\dot{Q}_{loss}| = 2.2 \text{ W}$$

Therefore, $|\dot{Q}_{loss}| = 326.8 \pm 2.2 \text{ W}$

6.4 APPENDIX D:

Uncertainty calculations for $\dot{Q}_{rh}$:

$$\delta\dot{Q}_{rh} = \sqrt{\left(\frac{\partial\dot{Q}_{rh}}{\partial\dot{m}_4} \delta\dot{m}_4\right)^2 + \left(\frac{\partial\dot{Q}_{rh}}{\partial\omega_4} \delta\omega_4\right)^2 + \left(\frac{\partial\dot{Q}_{rh}}{\partial h_4} \delta h_4\right)^2 + \left(\frac{\partial\dot{Q}_{rh}}{\partial h_3} \delta h_3\right)^2}$$

$$\frac{\partial\dot{Q}_{rh}}{\partial\dot{m}_4} = \frac{h_4 - h_3}{1 + \omega_4} = 0.92447$$

$$\frac{\partial\dot{Q}_{rh}}{\partial\omega_4} = -\frac{\dot{m}_4(h_4 - h_3)}{(1 + \omega_4)^2} = -0.17506$$

$$\frac{\partial \dot{Q}_{rh}}{\partial h_4} = \frac{\dot{m}_4}{1 + \omega_4} = 0.18912$$

$$\frac{\partial \dot{Q}_{rh}}{\partial h_3} = -\frac{\dot{m}_4}{1 + \omega_4} = -0.18912$$

$$\delta \dot{Q}_{rh} = \sqrt{(0.92447 * 0.002348)^2 + (-0.17506 * 0.00086856)^2 + (0.18912 * 2.2152)^2 + (-0.18912 * 1.0557)^2}$$

$$= 0.178 \pm 0.0464 \text{ kW}$$

6.5 APPENDIX E:

Uncertainty for $\dot{m}_{con}$:

As the uncertainty for v_{con} and t are zero, the equation simplifies to:

$$\delta \dot{m}_{con} = \sqrt{\left(\frac{\partial \dot{m}_{con}}{\partial V} \delta V\right)^2}$$

$$\frac{\partial \dot{m}_{con}}{\partial V} = \frac{1}{v_{con} * t} = 5.539$$

$$\delta \dot{m}_{con} = \sqrt{(5.539 * (0.00000849))^2} = 0.0000470$$

Therefore, $\dot{m}_{con} = 0.0003052 \pm 0.00004700 \text{ kg/s}$

6.6 APPENDIX F:

Uncertainty for $\dot{m}_{v3}$:

$$\delta \dot{m}_{v3} = \sqrt{\left(\frac{\partial \dot{m}_{v3}}{\partial \dot{m}} \delta \dot{m}\right)^2 + \left(\frac{\partial \dot{m}_{v3}}{\partial \omega_3} \delta \omega_3\right)^2}$$

$$\frac{\partial \dot{m}_{v3}}{\partial \dot{m}} = \frac{\omega_3}{1 + \omega_3} = 0.019127$$

$$\frac{\partial \dot{m}_{v3}}{\partial \omega_3} = \frac{\dot{m}}{(1 + \omega_3)^2} = 0.18499$$

$$\delta \dot{m}_{v3} = \sqrt{(0.019127 * 0.002348)^2 + (0.18499 * 0.000349)^2} = 0.000079$$

Therefore, $\dot{m}_{v3} = 0.00368 \pm 0.000079 \text{ kg/s}$

6.7 APPENDIX G:

Uncertainty for $\dot{m}_{v2}$:

METHOD 1:

$$\delta \dot{m}_{v2} = \sqrt{\left(\frac{\partial \dot{m}_{v2}}{\partial \dot{m}_{v3}} \delta \dot{m}_{v3}\right)^2 + \left(\frac{\partial \dot{m}_{v2}}{\partial \dot{m}_{con}} \delta \dot{m}_{con}\right)^2}$$

Where both partial expressions = 1

$$\delta \dot{m}_{v2} = \sqrt{(1 * 0.000079)^2 + (1 * 0.000047)^2} = 0.00009192$$

METHOD 2:

$$\delta \dot{m}_{v3} = \sqrt{\left(\frac{\partial \dot{m}_{v2}}{\partial \dot{m}} \delta \dot{m}\right)^2 + \left(\frac{\partial \dot{m}_{v2}}{\partial \omega_2} \delta \omega_2\right)^2}$$

$$\frac{\partial \dot{m}_{v2}}{\partial \dot{m}} = \frac{\omega_2}{1 + \omega_2} = 0.020184$$

$$\frac{\partial \dot{m}_{v2}}{\partial \omega_2} = \frac{\dot{m}}{(1 + \omega_2)^2} = 0.18461$$

$$\delta \dot{m}_{v3} = \sqrt{(0.020184 * 0.002348)^2 + (0.18461 * 0.00019982)^2} = 0.0000601$$

Therefore, using method 2 as the superior method:

$$\dot{m}_{v2} = 0.00388 \pm 0.0000601 \text{ kg/s}$$

6.8 APPENDIX H:

Uncertainty for $\dot{m}_a$:

$$\delta \dot{m}_a = \sqrt{\left(\frac{\partial \dot{m}_a}{\partial \dot{m}} \delta \dot{m}\right)^2 + \left(\frac{\partial \dot{m}_a}{\partial \omega_3} \delta \omega_3\right)^2}$$

$$\frac{\partial \dot{m}_a}{\partial \dot{m}} = \frac{1}{1 + \omega_3} = 0.98087$$

$$\frac{\partial \dot{m}_{v3}}{\partial \omega_3} = -\frac{\dot{m}}{(1 + \omega_3)^2} = -0.18499$$

$$\delta \dot{m}_a = \sqrt{(0.98087 * 0.002348)^2 + (-0.18499 * 0.00034905)^2} = 0.002304$$

Therefore, $\dot{m}_a = 0.1886 \pm 0.002304 \text{ kg/s}$

6.9 APPENDIX I:

Uncertainty for h_{con} :

At $T = 24^\circ\text{C}$, $h_{con} = h_f = 100.7 \text{ kJ/kg}$

At $T = 26^\circ\text{C}$, $h_{con} = h_f = 109.0 \text{ kJ/kg}$

Therefore, $Unc. = \frac{109.0-100.7}{2} = \pm 4.15 \text{ kJ/kg}$ and $h_{con} = 104.8 \pm 4.15 \text{ kJ/kg}$

6.10 APPENDIX J:

Uncertainty for $\dot{Q}_c$:

$$\frac{\partial \dot{Q}_c}{\partial \dot{m}_a} = h_3 - h_2 = -13.65$$

$$\frac{\partial \dot{Q}_c}{\partial h_3} = \dot{m}_a = 0.1886$$

$$\frac{\partial \dot{Q}_c}{\partial h_4} = -\dot{m}_a = -0.1886$$

$$\frac{\partial \dot{Q}_c}{\partial h_{con}} = \dot{m}_{con} = 0.0003052$$

$$\frac{\partial \dot{Q}_c}{\partial \dot{m}_{con}} = h_{con} = 104.8$$

$$\begin{aligned} \delta \dot{Q}_c &= \sqrt{(-13.65 * 0.002304)^2 + (0.1886 * 1.0557)^2 + (-0.1886 * 2.3675)^2 + (0.0003052 * 4.15)^2 + (104.8 * 0.0003052)^2} \\ &= 0.490 \end{aligned}$$

Therefore, $|\dot{Q}_c| = 2.54 \pm 0.490 \text{ kW}$

6.11 APPENDIX K:

Uncertainty for h_w :

At 23°C , $h_w = h_f = 96.47 \text{ kJ/kg}$

At 25°C , $h_w = h_f = 104.8 \text{ kJ/kg}$

Therefore, $Unc. = \frac{104.8-96.47}{2} = \pm 4.165 \text{ kJ/kg}$ and $h_w = 100.7 \pm 4.165 \text{ kJ/kg}$

6.12 APPENDIX L:

Uncertainty for $\dot{Q}_{ph} + \dot{Q}_b$:

$$\frac{\partial(\dot{Q}_{ph} + \dot{Q}_b)}{\partial \dot{m}} = 30.842$$

$$\frac{\partial(\dot{Q}_{ph} + \dot{Q}_b)}{\partial h_2} = 0.1901$$

$$\frac{\partial(\dot{Q}_{ph} + \dot{Q}_b)}{\partial h_1} = -0.1901$$

$$\frac{\partial(\dot{Q}_{ph} + \dot{Q}_b)}{\partial \dot{m}_{v2}} = -100.7$$

$$\frac{\partial(\dot{Q}_{ph} + \dot{Q}_b)}{\partial \dot{m}_{v1}} = 100.7$$

$$\frac{\partial(\dot{Q}_{ph} + \dot{Q}_b)}{\partial h_w} = -0.001675$$

$$\frac{\partial(\dot{Q}_{ph} + \dot{Q}_b)}{\partial \omega_1} = -5.862$$

$$\frac{\partial(\dot{Q}_{ph} + \dot{Q}_b)}{\partial \dot{Q}_{con}} = 1$$

$$\frac{\partial(\dot{Q}_{ph} + \dot{Q}_b)}{\partial \dot{W}_f} = -1$$

$$\begin{aligned} \delta(\dot{Q}_{ph} + \dot{Q}_b) = & [(30.842 * 0.002348)^2 + (0.0001496 * -5.862)^2 + (2.3675 * 0.1901)^2 \\ & + (0.8629 * -0.1901)^2 + (0.0000601 * -100.7)^2 + (0.000037 * 100.7)^2 \\ & + (-0.001675 * 4.165)^2 + (1 * 0.0022)^2 + (0)^2]^{1/2} = 0.4846 \end{aligned}$$

Therefore, $(\dot{Q}_{ph} + \dot{Q}_b) = 5.899 \pm 0.4846 \text{ kW}$